19 Show that $12^{-\frac{3}{2}}$ can be written in the form $\frac{\sqrt{3}}{h}$ where h is an integer.

$12^{\frac{-3}{2}} = \frac{1}{12\sqrt{12}} = \frac{1}{24\sqrt{3}}$

1 mark
Negative index: take reciprocal.

$\sqrt{12} = \sqrt{(4 \times 3)} = 2\sqrt{3}$

1 mark
Simplify √12.

$\frac{1}{24\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{\sqrt{3}}{24 \times 3}$

1 mark
Rationalise.

$= \frac{\sqrt{3}}{72}$ so $h = 72$

1 mark
√3/72.

(Total for Question 19 is 4 marks)